CS 6762
Fall 2023
SP, ML and FC
Homework 5

Answer all questions. Work alone. Total 100 points. Type or write neatly.

1. What are the main problems with each of the below: (10 points)
   a. Using P control by itself?
   b. Using I control by itself?
   c. Using D control by itself?

2. What is the difference between pole placement design and root locus? (5 points)

3. Consider a system with transfer function G(z) = Y(z)/U(z) = 0.5/(z-0.75) and proportional control. Find the range of values of K(p) such that the closed loop system (15 points)
   a. Is stable
   b. Has steady state error e(ss) to a unit step reference less than 0.1
   c. Has settling time k(s) < 10
   d. Has maximum overshoot M(p) < 0.1
   Note that it may not be possible to satisfy all criteria.

4. What is the difference between deciding on a PI controller, installing it in your system and evaluating the coefficients experimentally versus designing a PI controller using pole placement? (5 points)

5. What are the transfer functions for an Integral controller and for a PD controller? (10 points)

6. What are the main steps for developing PI control by pole placement? (5 points)

7. Sketch out the code (i.e., enumerate the main instructions required and pseudo code is fine) that you would need to implement a PID controller on a microprocessor. (10 points)

8. Consider the IBM Lotus Domino server with G(z) = Y(z)/U(z) = (.47)/(z-.43) and with a PI controller. Let Kp = 2 and Ki = 1. Find the closed loop transfer function Fr(z) = Y(z)/R(z). Also assume that there is no transducer. (10 points)

9. What computing system issues are not typically accounted for in classical feedback control theory? (5 points)

10. Go to the WEB and investigate reinforcement learning. Write a 3⁄4 to 1page answer that summarizes of the main ideas of reinforcement learning AND compares reinforcement learning to BOTH deep neural networks and feedback control. (20 points)